Paediatric Endoscopy, Part Two: the Biopsy Protocol.

Imagine a twelve-year-old boy who has had trouble swallowing meat for a year. At endoscopy his oesophagus looks completely normal: no rings, no furrows, no white exudates. The endoscopist takes two biopsies from the lower oesophagus, puts them in one jar with the gastric and duodenal samples, and writes "dysphagia" on the form. The report comes back as mild reflux changes. Six months later, at a second endoscopy and a second general anaesthetic, a colleague takes six biopsies from the upper and lower oesophagus in two labelled jars, and the peak count is forty eosinophils per high-power field. The boy had eosinophilic oesophagitis the whole time. Nothing was wrong with the pathologist, and nothing was wrong with the endoscope. What was wrong was the biopsy protocol.

The clinical question of this chapter is how many biopsies to take, from where, and in which containers, for the common diagnostic questions in paediatric endoscopy. The main idea is this. The diseases we look for, eosinophilic oesophagitis, coeliac disease, inflammatory bowel disease and Helicobacter pylori infection, are all patchy, and a biopsy forceps samples only a few square millimetres of mucosa. So the diagnostic yield is decided by the endoscopist before the pathologist ever sees a slide. The protocols are simple to remember: at least six oesophageal biopsies from two levels; at least four biopsies from the distal duodenum plus at least one from the bulb; at least two biopsies from every segment of the ileum and colon, plus the upper tract, in suspected inflammatory bowel disease; and six gastric biopsies, three from the antrum and three from the corpus, when Helicobacter pylori is suspected. The hard part is not remembering them. The hard part is doing them every time, including when the mucosa looks normal.

We will move through the topic in order. First, why sampling matters, disease by disease. Second, the oesophagus, and eosinophilic disease beyond it. Third, the duodenum and coeliac disease. Fourth, the ileum and colon in inflammatory bowel disease. Fifth, the stomach and Helicobacter pylori, and a few special requests. Sixth, specimen handling and the quality standards. We finish with what the guidelines agree and disagree on, and with what to do differently at your next list.

A note on sources. For eosinophilic oesophagitis we use the 2018 international AGREE consensus on diagnostic criteria, led by Dellon; the 2025 clinical guideline of the American College of Gastroenterology, the ACG, also led by Dellon; and the 2024 update from the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, ESPGHAN, led by Amil-Dias. For coeliac disease we use the 2020 ESPGHAN guideline led by Husby and the 2023 ACG guideline led by Rubio-Tapia. For inflammatory bowel disease we use the revised Porto criteria of 2014, led by Levine. For Helicobacter pylori we use the joint 2023 guideline of ESPGHAN and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, NASPGHAN, led by Homan. We also draw on the 2021 tissue sampling guideline of the European Society of Gastrointestinal Endoscopy, ESGE, and on the 2022 quality standards of the Paediatric Endoscopy Quality Improvement Network, known as PEnQuIN.

Part one. Patchy disease, tiny samples.

Each disease needs its own sampling rule because each is distributed differently in the gut.

Eosinophilic oesophagitis is patchy along the whole length of the oesophagus, and the eosinophil count can differ a great deal from one fragment to the next, even a few centimetres apart. That is why the number of biopsies and the number of levels matter more than how the mucosa looks.

Coeliac disease causes patchy villous atrophy, and in a small number of children the damage is found only in the duodenal bulb. That is why we take both distal duodenal and bulb biopsies, and why orientation matters.

Crohn disease is famous for skip lesions, and its most specific microscopic feature, the granuloma, is focal and often sits in mucosa that looks normal. That is why every segment is sampled, including the upper tract.

Ulcerative colitis in children is more often patchy, and more often spares the rectum, than in adults. That is why a full colonoscopy with segmental biopsies is needed, not just a look at the rectum.

Helicobacter pylori infection is patchy as well, and when a child is taking acid suppression, the organisms tend to move from the antrum up into the corpus. That is why both the antrum and the corpus are sampled, and why tissue is taken for culture or molecular testing as well as for histology.

How often does normal-looking mucosa hide disease in children? A study by Sheiko and colleagues, published in 2015, compared the endoscopic and histological findings of one thousand paediatric upper endoscopies in Colorado. Thirty percent of children whose biopsies showed eosinophilic oesophagitis, and forty-three percent of those with coeliac disease, had mucosa that looked normal to the endoscopist. If only visibly abnormal areas had been biopsied, forty-eight and a half percent of histological disease would have been missed. Normal mucosa is not a reason to skip a biopsy. It is the reason the protocol exists.

Part two. The oesophagus: six biopsies approach certainty.

The best way to understand the six-biopsy rule is to look at how sensitivity grows with each biopsy. In an adult study published by Gonsalves and colleagues in 2006, a single biopsy detected eosinophilic oesophagitis, defined as at least fifteen eosinophils per high-power field, in only fifty-five percent of patients, while five biopsies detected it in all of them. A single-centre series of one hundred and two consecutive cases, published by Nielsen and colleagues in 2014, found a sensitivity of about sixty-three percent with one fragment, ninety-five percent with three, ninety-eight percent with four, and just over ninety-nine percent with five. The authors advised at least four fragments, and said that more than six were not needed. When these two studies are drawn as a bar chart, the curve rises steeply over the first three biopsies and then flattens close to one hundred percent. If we also want to see fibrosis of the lamina propria, which needs deeper tissue, about seven or more biopsies from the middle and lower oesophagus have been reported to detect it in more than ninety-five percent of cases.

What do the guidelines ask for? The 2018 AGREE consensus defined the diagnosis as symptoms of oesophageal dysfunction, at least fifteen eosinophils per high-power field, which is about sixty eosinophils per square millimetre, and exclusion of other causes. It also removed the old requirement for a trial of a proton pump inhibitor. The 2025 ACG guideline makes a strong recommendation to take at least six biopsies from at least two levels of the oesophagus, which in practice means two to four from the upper half and two to four from the lower half, aiming at furrows and exudates when they are present. It also recommends biopsy at the time of a food bolus impaction, which can save a second anaesthetic. The ESGE tissue sampling guideline of 2021 adds that the proximal and distal biopsies should be placed in separate containers. The ESPGHAN 2024 update for children uses the same threshold of fifteen eosinophils per high-power field.

One point of citation is worth getting right. The 2020 guideline of the American Gastroenterological Association and the Joint Task Force on allergy practice parameters is a management guideline. It does not set a diagnostic biopsy number, so the source for the six-biopsy rule is the ACG guideline, with ESGE and ESPGHAN in agreement.

What should we ask the pathologist to report? Not just a yes or no. We want the peak eosinophil count per high-power field, the area of the field the pathologist used, and, ideally, the other features scored in the Eosinophilic Oesophagitis Histology Scoring System, published by Collins and colleagues in 2017. That system grades and stages eight features: eosinophil density, basal zone hyperplasia, eosinophil abscesses, surface layering, dilated intercellular spaces, surface epithelial alteration, dyskeratotic cells, and lamina propria fibrosis.

At the first paediatric endoscopy, biopsies from the stomach and duodenum are taken as well, because eosinophilic disease may extend beyond the oesophagus, and because other diagnoses can present with the same symptoms.

Part three. Normal eosinophils are not zero.

Outside the oesophagus, eosinophils are a normal resident of the gut wall, and the normal numbers vary by segment. Thresholds proposed by Collins and colleagues in 2018 make this clear. Eosinophilic gastritis is suggested by at least thirty eosinophils per high-power field in at least five fields, or at least seventy per field in at least three fields. In the duodenum the proposed threshold is more than fifty-two per field, in the terminal ileum more than fifty-six, in the caecum and right colon more than one hundred, in the transverse and left colon more than eighty-four, and in the rectosigmoid more than sixty-four. These values are roughly twice the normal peak counts. Notice that the right colon normally has the highest counts. The 2024 joint ESPGHAN and NASPGHAN guideline on eosinophilic disease beyond the oesophagus, led by Papadopoulou, gives organ-specific thresholds and asks for specimens labelled by site. The practical lesson is simple: without the segment written on the jar, an eosinophil count cannot be interpreted.

Part four. The duodenum: four biopsies, then the bulb.

For coeliac disease, the number of duodenal biopsies changes the diagnosis rate dramatically. In a study by Pais and colleagues in 2008, two biopsies confirmed the diagnosis in ninety percent of

patients with coeliac disease, and four biopsies confirmed it in all of them. In a large American pathology database study published by Lebwohl and colleagues in 2011, which included one hundred and thirty-two thousand three hundred and fifty-two patients, only thirty-five percent had at least four specimens taken. The coeliac diagnosis rate was one point eight percent when at least four specimens were taken, and only zero point seven percent when fewer were taken, about two and a half times lower. Among patients biopsied specifically for suspected coeliac disease or malabsorption, taking at least four specimens raised the odds of diagnosis about sevenfold. A later study by Shamban and colleagues in 2017 found a yield of zero point three percent with fewer than four specimens, four point seven percent with four or more, and twelve point eight percent with six. In that study, paediatric gastroenterologists took more samples than other endoscopists. The lesson is that below four specimens, the missed diagnoses are not a small fraction. They are most of the diagnoses.

What about the bulb? In a multicentre Italian study of six hundred and sixty-five children with coeliac disease, published by Bonamico and colleagues in 2008, two point four percent, sixteen children, had lesions only in the duodenal bulb. That is why the ESPGHAN 2020 guideline asks for at least four biopsies from the distal duodenum and at least one from the bulb, taken while the child is still eating gluten. The 2023 ACG guideline asks for four distal biopsies plus one or two from the bulb, which is the same in practice.

Orientation is the next problem. The pathologist judges villous atrophy by comparing the height of the villi with the depth of the crypts, and a villous to crypt ratio below two indicates a lesion. That comparison only works if the specimen is cut at right angles to the surface. ESPGHAN 2020 says that only optimally oriented biopsies should be read, and that the Marsh–Oberhuber grade can only be given when orientation is correct. In the data behind the guideline, only about half of paediatric biopsies were optimally oriented. In the ProCeDE study led by Werkstetter, local and central pathologists disagreed on whether the child had coeliac disease in seven percent of cases, and on the Marsh grade in fifty-eight percent. When histology and serology disagree, the guideline recommends recuts of the block or a second pathologist's opinion.

There is one simple technique that helps. In an adult study by Latorre and colleagues in 2015, taking a single bite with each pass of the forceps gave well-oriented specimens in sixty-six percent, compared with forty-two percent when two bites were taken per pass. So the practical rule is: one bite per pass, the bulb in its own jar, and the serology result written on the requisition.

Part five. Children are under-sampled too.

It would be comforting to think that paediatric endoscopists do better than adult ones. A study of American children by Wallach and colleagues, published in 2017, suggests otherwise. Among nine thousand one hundred and seventy-one children who had duodenal biopsies, only thirty-five percent had at least four. Coeliac disease was found in five percent of children whose biopsies followed the guideline, and in only zero point seven percent of those whose biopsies did not, an adjusted odds ratio of about six. Among eight thousand two hundred and eighty children who had oesophageal biopsies, only about eight percent had biopsies from two specified levels, and

sixty-eight percent of specimens came from a site that was not even labelled. Eosinophilic oesophagitis was found in twenty-five and a half percent when the protocol was followed, compared with about eight percent when it was not, an adjusted odds ratio of about two and a half. In both diseases, detection rose steadily with the number of biopsies.

Part six. The ileum and colon: ten segments, labelled jars.

For suspected inflammatory bowel disease, the revised Porto criteria of 2014 recommend an upper endoscopy and an ileocolonoscopy in every child, plus imaging of the small bowel, usually by magnetic resonance enterography or capsule, unless the picture is typical ulcerative colitis. Biopsies are taken from every segment, whether or not it looks inflamed. In practice this means ten sites: the oesophagus, the stomach, the duodenum, the terminal ileum, the caecum, the ascending colon, the transverse colon, the descending colon, the sigmoid colon and the rectum. A 2018 position paper of the Porto group, led by Oliva, repeats the call for multiple biopsies from each segment even without visible disease, and the ESGE guideline of 2021 asks for at least two biopsies per segment in separate containers. The reason for separate jars is that the distribution of inflammation is itself the diagnosis. Continuous colitis from the rectum upwards suggests ulcerative colitis, while patchy disease, ileal involvement or focal gastritis suggests Crohn disease. One jar for the whole colon throws that information away.

What does the upper endoscopy add? In a Dutch study of seventy children with Crohn disease, published by Hummel and colleagues in 2012, upper tract findings established or changed the diagnosis in eleven percent. Granulomas were found in the upper tract in thirty percent, nineteen of twenty-one of them in the stomach, and duodenal inflammation was seen in nineteen percent of children with Crohn disease, compared with none of those with ulcerative colitis. In a larger study by De Matos and colleagues in 2008, granulomas were found in sixty-one percent of untreated children, compared with about a quarter of children who had already been treated. So the diagnostic endoscopy, before steroids, is the best chance to find them. And yet, in a Hungarian national paediatric registry reported by Müller and colleagues in 2021, upper tract biopsies were taken in only sixty-three percent of upper endoscopies done for suspected inflammatory bowel disease.

Two places change the label more often in children than in adults. The first is the terminal ileum. About one child in ten with Crohn disease has no colonic involvement at all, so isolated ileitis will be missed unless the ileum is reached and biopsied. PEnQuIN sets a target terminal ileal intubation rate of at least eighty-five percent. In the Hungarian registry, ileoscopy rose from fifty-two percent to eighty-four percent between 2007 and 2016. The second is the rectum. In a study by Glickman and colleagues in 2004, rectal sparing was seen in about thirty percent of untreated children with ulcerative colitis, compared with about three percent of adults. So a sigmoidoscopy is not enough for the first diagnosis. And rectal sparing or patchy colitis in an untreated child does not by itself mean Crohn disease; under the Porto criteria, atypical ulcerative colitis is still ulcerative colitis.

Part seven. The stomach: six gastric biopsies, two purposes.

The 2023 joint ESPGHAN and NASPGHAN guideline on Helicobacter pylori in children, published in 2024, recommends that diagnosis be based on biopsies, using culture or a molecular test together with histology scored by the updated Sydney system. It makes a strong recommendation to take at least six gastric biopsies, three from the antrum and three from the corpus. Two from the antrum and two from the corpus, from the anterior and posterior walls, go for histology. One from the antrum and one from the corpus go for culture, polymerase chain reaction or a rapid urease test. The tissue for culture should be taken before the forceps has touched formalin. The guideline recommends real-time polymerase chain reaction to detect resistance to clarithromycin.

Infection is confirmed by a positive culture or molecular test, or by positive histology together with a positive rapid urease test. Active bleeding and acid suppression reduce the sensitivity of all of these. Because the guideline does not support a test-and-treat strategy in children, the endoscopy is the place where the decision about treatment is made. So if we are going to treat Helicobacter pylori in a child, we should take tissue that tells us which antibiotic will work.

Part eight. When the question changes the handling.

A few indications need different handling, and the fixative must be chosen before the procedure, not after the specimen is already in formalin.

For disaccharidase activity, for example when a toddler has watery diarrhoea after sucrose and we suspect sucrase-isomaltase deficiency, at least two extra biopsies are taken from the distal duodenum and snap-frozen immediately, with no formalin, no saline and no filter paper.

For microscopic colitis, which is mainly an adult disease, the ESGE guideline recommends two biopsies from the right colon in one jar and two from the left colon in another, because the mucosa looks normal.

For graft-versus-host disease after stem cell transplantation, an adult series by Aslanian and colleagues in 2012 found that a rectal biopsy alone detected eighty-nine percent of cases, and that normal-looking mucosa was often positive. A small paediatric series of twenty procedures found a sensitivity of eighty percent in the duodenum, but only twenty-two percent in the oesophagus and thirty percent in the stomach. So the rectosigmoid and the duodenum are the most useful sites.

And for a child with a food bolus impaction, oesophageal biopsies at the time of disimpaction may make the diagnosis of eosinophilic oesophagitis and avoid a second anaesthetic.

Part nine. From forceps to report.

Between the biopsy and an interpretable slide there are five steps. First, the bite: one specimen per pass when orientation matters, avoiding crush. Second, orientation: the mucosal side up, and duodenal specimens laid flat. Orientation on filter paper is expert practice rather than guideline-level evidence, but the evidence that poor orientation causes disagreement is strong. Third, separation: one labelled jar per site or level, with fresh tissue taken first. Fourth, the request: the clinical question, the serology, whether the child is eating gluten, the medications, and a

request for an eosinophil count when relevant. Fifth, discussion: recuts or a second opinion when the histology and the clinical picture disagree. A well-taken biopsy with no clinical question attached is only half a test.

Part ten. What PEnQuIN measures, and what it does not.

The PEnQuIN quality standards, published in 2022 by Walsh and colleagues, reached international consensus on forty-nine standards and forty-seven indicators for paediatric endoscopy. Among them are a caecal intubation rate of at least ninety percent, a terminal ileal intubation rate of at least eighty-five percent, and adequate bowel preparation in at least eighty percent of colonoscopies. PEnQuIN also asks that biopsies be taken for appropriate indications. But the number of biopsies per site did not reach consensus as a quality indicator, so PEnQuIN sets no number for eosinophilic oesophagitis, coeliac disease or inflammatory bowel disease. It leaves the counts to the disease-specific guidelines. That means we have to audit our own counts against those guidelines; the quality framework will not do it for us.

Part eleven. What the guidelines say.

North American and European documents agree on most of the protocol. For the number of oesophageal biopsies, the ACG asks for at least six from at least two levels, and the European documents ask for the same, two to four proximal and two to four distal. They are aligned. For containers, ESGE asks for proximal and distal oesophageal biopsies in separate jars, while the ACG states no such requirement. They differ. For coeliac biopsies, the ACG asks for four distal plus one or two bulb biopsies and ESPGHAN for at least four distal plus at least one bulb biopsy. Aligned. For Helicobacter pylori, a single joint guideline covers both continents. Aligned. For inflammatory bowel disease, both sides ask for upper and lower endoscopy with every segment sampled. Aligned. And neither side has a quality indicator for biopsy counts, which is a gap.

The material difference is whether a child with suspected coeliac disease is biopsied at all. ESPGHAN 2020 allows a diagnosis without biopsy when the tissue transglutaminase IgA antibody is at least ten times the upper limit of normal and endomysial antibodies are positive in a second sample, whereas North American practice still generally advises biopsy. But once the endoscope is in, both sides ask for the same protocol.

Part twelve. What to do differently at your next list.

Let us go back to the boy with dysphagia. At his first endoscopy, the right plan was at least six biopsies from the proximal and distal oesophagus in separate labelled jars, plus biopsies from the stomach and duodenum, with a request for the peak eosinophil count, whatever the mucosa looked like. A seven-year-old with a tissue transglutaminase three times the upper limit of normal needs at least four distal duodenal biopsies and at least one from the bulb, taken with a single bite per pass, while still on gluten. A fourteen-year-old with diarrhoea and a faecal calprotectin of nine hundred micrograms per gram needs an ileocolonoscopy and an upper endoscopy, with at least two biopsies from every segment, labelled, whether or not the mucosa looks inflamed. A ten-year-old with a nodular antrum and suspected Helicobacter pylori needs three antral and three corpus biopsies,

with tissue for culture or polymerase chain reaction to guide antibiotic choice. A toddler with watery diarrhoea after sucrose needs extra distal duodenal biopsies snap-frozen for disaccharidase activity. And a child with diarrhoea after a stem cell transplant needs rectosigmoid and duodenal biopsies, including normal mucosa.

The whole chapter can be held in one sentence. Name the question, count the biopsies, and label the jars.

Questions to test yourself.

Question one. How many oesophageal biopsies are recommended to diagnose eosinophilic oesophagitis, and from where? Answer: at least six, from at least two levels, two to four from the proximal and two to four from the distal oesophagus, ideally in separate containers.

Question two. What was the sensitivity of a single oesophageal biopsy in the Gonsalves and Nielsen studies? Answer: about fifty-five percent in the adult Gonsalves study and about sixty-three percent in the Nielsen series.

Question three. What is the coeliac biopsy protocol in ESPGHAN 2020? Answer: at least four biopsies from the distal duodenum and at least one from the bulb, on a gluten-containing diet, read only when optimally oriented.

Question four. What happened to the coeliac diagnosis rate when fewer than four specimens were taken in the Lebwohl study? Answer: it fell from one point eight percent to zero point seven percent, about two and a half times lower.

Question five. In American children, how often were the coeliac and eosinophilic oesophagitis protocols followed? Answer: about thirty-five percent had at least four duodenal biopsies, and only about eight percent had oesophageal biopsies from two specified levels.

Question six. Why must the terminal ileum and the rectum both be examined in a child with suspected inflammatory bowel disease? Answer: about one child in ten with Crohn disease has no colonic involvement, and about thirty percent of untreated children with ulcerative colitis have rectal sparing.

Question seven. How should gastric biopsies be allocated when Helicobacter pylori is suspected? Answer: at least six, two antrum and two corpus for histology, and one antrum and one corpus for culture, polymerase chain reaction or rapid urease testing, taken before the forceps touches formalin.

Question eight. Does PEnQuIN set a number of biopsies per site? Answer: no. Biopsy counts did not reach consensus as a quality indicator, so the counts come from the disease-specific guidelines.

End of chapter.